DEVELOPMENT OF RES4 SMART CONTRACT

The data defined is mostly for managing the various attributes of the tokens. The functions are developed in four major categories, as delineated by comments in listing 9.1:

- Mapping for various attributes
- Functions, events, and data for ERC721 compliance
- Application-specific (RES4-specific) functions
- Internal functions to support these and other utility functions
- You'll have to implement all the functions for ERC721 compliance, but your Dapp may not need or use all of them. That's why you see two sections in listing 9.1: one for the ERC721 functions required for the RES4 Dapp and one at the bottom that has ERC721 functions used by the RES4 token Dapp but implemented for compliance. You can find the complete smart contract in the codebase of this chapter.

Listing 9.1 RES4 smart contract (RES4.sol)

```
pragma soldity >=0.4.22 <=0.6.0;
import "./helper_contracts/ERC721.sol";

contract RES4 is ERC721 {                                          ❶
    struct Asset{
        uint256 assetId;
        uint256 price;
    }

    uint256 public assetsCount;
    mapping(uint256 => Asset) public assetMap;                     ❷
    address public supervisor;                                     ❷
    mapping (uint256 => address) private assetOwner;               ❷
    mapping (address => uint256) private ownedAssetsCount;         ❷
    mapping (uint256 => address) public assetApprovals;            ❷


//Events
    event Transfer(address from, address to, uint256 tokenId);     ❸
```